

THE WEIL EXPLICIT FORMULA FOR A CONDUCTOR-27 CM CURVE: A NUMERICAL BALANCE OF ZEROS AGAINST EISENSTEIN PRIMES

ABSTRACT. For the CM elliptic curve $E : y^2 + y = x^3$ of conductor 27, with CM by the Eisenstein integers $\mathbb{Z}[\omega]$, we verify the Weil explicit formula numerically by computing its two sides independently. The spectral side sums a Gaussian test function over the nontrivial zeros of $L(E, s)$ on the critical line; the arithmetic side sums the von Mangoldt weights $\Lambda_f(p^k) = (\pi^k + \bar{\pi}^k) \log p$ built from the Frobenius traces $a_p = \pi + \bar{\pi}$, $\pi \in \mathbb{Z}[\omega]$, together with the conductor constant $\log(27/4\pi^2)$ and an archimedean digamma integral. The two sides, sharing no common input, agree to relative error 8.5×10^{-6} . This is the identity in which the two objects developed elsewhere in this program—the zeros of the CM L -function and the split primes of $\mathbb{Q}(\omega)$ —are exhibited as two encodings of one quantity.

1. THE IDENTITY

Let $E : y^2 + y = x^3$, the elliptic curve of conductor 27 with complex multiplication by $\mathbb{Z}[\omega]$, $\omega = e^{2\pi i/3}$. Its L -function $L(E, s) = \sum_{n \geq 1} a_n n^{-s}$ has completed form

$$\Lambda(s) = \left(\frac{\sqrt{27}}{2\pi} \right)^s \Gamma(s) L(E, s), \quad \Lambda(s) = \Lambda(2-s),$$

and equals the Hecke L -function $L(s, \psi)$ of a Grossencharacter ψ of $\mathbb{Q}(\omega)$; equivalently $L(E, s) = L(f, s)$ for the weight-2 level-27 newform $f = (\eta(3\tau)\eta(9\tau))^2$. For a prime $p \equiv 1 \pmod{3}$ (split in $\mathbb{Z}[\omega]$), $a_p = \pi + \bar{\pi}$ where π is the primary generator of the prime above p ; for $p \equiv 2 \pmod{3}$ (inert), $a_p = 0$.

The Weil explicit formula pairs a test function against the zeros and against the primes. For the Gaussian pair

$$g(u) = e^{-u^2/2\sigma^2}, \quad h(t) = \sigma\sqrt{2\pi} e^{-\sigma^2 t^2/2} = \widehat{g}(t),$$

and zeros $\rho = 1 + i\gamma_j$ of $L(E, s)$ on the critical line $\Re(s) = 1$, the identity reads

$$\underbrace{2 \sum_j h(\gamma_j)}_{\text{spectral side}} = \underbrace{\log \frac{27}{4\pi^2} + \frac{1}{\pi} \int_{-\infty}^{\infty} \Re \psi_0(1+it) h(t) dt - 2 \sum_{n \geq 2} \frac{\Lambda_f(n)}{n} g(\log n)}_{\text{arithmetic side}},$$

where ψ_0 is the digamma function and the von Mangoldt weights are powers of Eisenstein-prime traces,

$$\Lambda_f(p^k) = (\pi^k + \bar{\pi}^k) \log p = 2 \Re(\pi^k) \log p \quad (p \text{ split}),$$

extended by $\Lambda_f(p^k) = (\alpha_p^k + \beta_p^k) \log p$ with $\alpha_p + \beta_p = a_p$, $\alpha_p \beta_p = p$, in general.

2. INDEPENDENT COMPUTATION OF THE TWO SIDES

Spectral side. We formed $\Lambda(1+it)$ from the smoothed series in the coefficients a_n (computed from the η -product to $n < 3000$) and located its zeros by root-finding. The first few are

$$\gamma_j = 4.043, 6.049, 8.218, 9.429, 10.909, 12.716, \dots,$$

and we used 28 zeros up to $\gamma \approx 39.6$. The curve has rank 0 ($L(E, 1)/\Omega = 1/9 \neq 0$), so there is no zero at the centre $s = 1$.

Arithmetic side. The conductor constant is $\log(27/4\pi^2) = -0.3799\dots$. The archimedean term is the digamma integral above. The prime sum runs over prime powers, with $\Lambda_f(n)$ assembled from the Frobenius data: a_p from the newform, then α_p, β_p as the roots of $x^2 - a_p x + p$. No zero of $L(E, s)$ enters this side; it is built entirely from the split primes of $\mathbb{Z}[\omega]$ and their traces.

σ	spectral side	arithmetic side	relative error
0.40	0.6605213563	0.6605157570	8.5×10^{-6}
0.45	0.4892997096	0.4893025481	5.8×10^{-6}
0.50	0.3513144956	0.3514004826	2.4×10^{-4}

TABLE 1. The two sides, computed from disjoint inputs (a list of zeros; a list of primes), agree. Accuracy is best at small σ , where the finite zero list suffices; the growth of the residual with σ is truncation of the zero sum (the test function then weights zeros beyond those computed), not a failure of the identity.

Balance.

3. READING

Observation 1. The spectral and arithmetic sides share no input. One is the multiset of zeros of $L(E, s)$; the other is the multiset of split primes of $\mathbb{Q}(\omega)$ together with the traces $\pi^k + \bar{\pi}^k$. They agree to five significant figures because the explicit formula makes them the same number. This is the precise sense in which the zeros of the CM L -function and the arithmetic of $\mathbb{Z}[\omega]$ are two encodings of one object.

The constant $\log(27/4\pi^2)$ carries the conductor 27—the same 27 of the cubic-residue representation $4p = L^2 + 27M^2$ and of the curve’s minimal model. Every ingredient of the arithmetic side was established independently: the traces $a_p = \pi + \bar{\pi}$ as primary-generator traces in $\mathbb{Z}[\omega]$; the modularity $L(E, s) = L(f, s)$ via the η -product; the split/inert law from χ_3 . The explicit formula is where these become a single identity, and the identity balances.

Remark 2 (Scope). This is a numerical verification of a proved theorem (Weil’s explicit formula), specialised to one CM curve and carried to ~ 6 digits; it is a consistency check on the tower assembled in the companion notes, not a new theorem. It relates the zeros to the primes but says nothing about the *location* of the zeros (the Riemann Hypothesis for $L(E, s)$) or the *spacing* of the primes: the formula is the bridge between the two sides, not a determination of either.

RELATION TO THE COMPANION NOTES

This note closes a chain verified across the program: the split/inert law of $\mathbb{Z}[\omega]$ ($p \equiv 1, 5 \pmod{6}$, the character χ_3); the factorisation $\zeta_{\mathbb{Q}(\omega)} = \zeta \cdot L(s, \chi_3)$; the conductor-27 CM curve with $a_p = \pi + \bar{\pi}$ and cubic reciprocity $4p = L^2 + 27M^2$; modularity through $(\eta(3\tau)\eta(9\tau))^2$; the central value $L(E, 1) = \Omega/9$ and Chowla–Selberg period $\Omega = \sqrt{3}\Gamma(1/3)^3/2\pi$; and the CM (arcsine) Sato–Tate law. The Weil explicit formula is the identity in which the two ends of that chain—zeros and primes—meet.

REFERENCES

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